

Yoomin Lee

yoomin.lee.3@usc.edu • <https://yoominlee.github.io> • [LinkedIn](#)

RESEARCH INTERESTS

Passionate about advancing data driven architectural workflows by building computational systems that make design decisions faster, and more transparent. With a background spanning architecture and computer science, research interests focus on practical pipelines that connect parametric modeling and BIM with automated analysis, clear visual communication, and reproducible evaluation. Particular emphasis is placed on human centered tools that keep designers in control while responsibly embedding data and AI to improve reliability, reduce repetitive coordination work, and support scalable adoption across multidisciplinary teams.

EDUCATION

- 2025–** **University of Southern California** Los Angeles, CA
M.S. in Computer Science (AI track)
Grader - Data Science at Scale (DSCI 550)
- 2021** **State University of New York at Oswego** Oswego, NY
Exchange student (Fall semester)
- 2019–24** **Hanyang University** Seoul, Korea
B.S. in Interior Architectural Design, Computer Science
GPA 4.04/4.5 (Double Major)

SELECTED PROJECTS

- 2025–** **USC Ginsburg Hall Digital Twin (Living Lab)** USC
Advisor: Prof. David Gerber, Link: [Project page](#)
Developed a research-grade digital twin of USC Ginsburg Hall to enable standardized data, reproducible evaluation, and rapid prototyping with real-world building data.
Integrated multi-source IoT streams (temperature, occupancy, lighting) with 3D building geometry and implemented cloud-based ingestion (Azure Blob Storage) with real-time API access.
- 2025–** **Semi-Supervised Multi-View Truck Detection** USC
Advisor: Prof. Seon-Ho Kim
Training computer vision model in teacher-student pipeline with pseudo-label selection for cross-angle generalization; semi-supervised learning for domain shift in real world video.
Modular scripts for training, pseudo-label generation, and evaluation to enable fast iteration and reuse.
- 2024** **Adaptive Camera Sensor for Vision Models** Seoul National University
Advisor: Prof. Hyungsin Kim
Co-collected parameter-diverse image data; implemented acquisition/QA scripts for timestamp alignment, integrity checks, and duplication control.
Designed real-world evaluation using ImageNet-ES Natural to validate robustness under domain shift.

- 2023–24 Fine-Grained Recognition with Multimodal Context** Hanyang University
Advisor: Prof. Dong-Jin Kim
 Extended CLIP with geo/text context; curated metadata and ran ablations for prompt/context strategies.
 Maintained weekly SOTA reviews; emphasized rare-class robustness and clear documentation.
- 2023 Similar Furniture Classification** Hanyang University
Advisor: Prof. Jae-hyuk Cha
 Trained (fine-grained) ML vision model to classify similar furnitures. Generated multi-view/light renders from 3D models; built an end-to-end TensorFlow pipeline.
 Documented failure modes and kept class balance to support reliable baseline comparisons.
- 2023 LC-2K Assembler & Simulator** Hanyang University
Course Project, Link: [Assembler](#) · [Simulator](#)
 Assembler with label resolution, pseudo-instruction expansion, and error handling.
 Cycle-accurate simulator covering registers/memory/branches with PC/state trace export.
- 2022–23 Person-Following for Indoor Robots** COGA Robotics
Advisor: Jinhjung Kim
 Implemented ROS nodes with RGB-D/LiDAR calibration; conducted lighting/occlusion stress tests and runtime profiling on Jetson-class devices.
 Focused on stability, recovery behavior, and maintainable node-level interfaces for deployment.

TECHNICAL SKILLS

Languages	Python, C/C++, JAVA, PostgreSQL, L ^A T _E X
ML/CV	PyTorch, TensorFlow, OpenCV
Systems	Linux, Git, Docker, VS Code, CUDA
Data	NumPy, Pandas, Matplotlib; experiment logging and reproducible configs
Graphics	Autodesk AutoCAD, Figma, Adobe Photoshop, Adobe Illustrator, Adobe InDesign, Rhino/Grasshopper

INTERNSHIPS

- 2024 AIoT Group** Seoul National University
Research Member ([Link](#))
- 2023–24 Multimodal AI Lab** Hanyang University
Undergraduate Researcher ([Link](#))
- 2022–23 COGA Robotics** Seoul, Korea
Intern Researcher ([Link](#))

PREPRINTS & WORKING PAPERS

2025 Pre- and Post-Occupant Satisfaction in Modern Office Hot-Desking Systems:
A Comparative Study of Workplace Environments

Zhihong Liu, Yifan Zhang, Tsz Him Ian Chu, Ghazal Chizarian, Joon Ho Choi, and Yoomin Lee

Under review

DESIGN WORKS & EXHIBITIONS

2023 Urban Aquaculture

Hanyang University

Interior Architecture Design Studio 2 ([Link](#))

2022 Infectious Disease Response HUB

Hanyang University

Thesis Project Capstone Design ([Link](#))

POSTER PRESENTATIONS

2025 Multi-View Truck Detection via Semi-Supervised Learning - [KOCSEA](#)

DATASETS

2024 ImageNet-ES

Seoul National University

Co-collected ([Link](#))

Real-world image data under varied parameters for robustness for ML models.

SELECTED COURSEWORK

Graduate Machine Learning, Information Retrieval, High-Performance Computing

Undergraduate Computer Vision, Deep Learning Methods and Applications, Data Structures, Probability & Statistics, VR AR Design, Database Systems, Computer Architecture, Micro-Processor Applications, Object-Oriented Systems Design

TEACHING & MENTORING

2025 Fall Grader, Data Science at Scale (DSCI 550)

USC

2025 Fall Viterbi Mentor

USC

2025 Summer Student Coordinator, USC Integrated Media Systems Center Hackathon

Link: [Homepage](#), [News](#)

AWARDS & HONORS

- 2024 SNU Ambient AI Competition** Seoul National University
1st Prize, Link: [Youtube](#), [Github](#)
Created on-device real-time blurring application. Team of 4
- 2020 DB GAPS Challenge**
2nd Place, DB Kim Jun Ki Cultural Foundation
Used financial and market data to build automated investment strategy with team of 4

EXTRACURRICULAR ACTIVITIES

- 2019–24 Hanyang University Hardware & Software Club** Hanyang University
President (2022), Vice President (2021)
HUHS, a central club at Hanyang University with about 200 members.
Planned and conducted various events in addition to seminars, founding ceremonies, and studies.
- 2022–24 Hanyang University Judo Team** Hanyang University
Member
Regular team training while balancing coursework; developed discipline, resilience, and time management.
- 2020 Vietnam Volunteer Service, Together Hanyang** Tây Ninh, Vietnam
Art team leader, Children's education team
'Together Hanyang', a foundation under Hanyang University. Overseas volunteer, worked as the art team leader of the children's education team.
- 2019 Student Council (College of Human Ecology)** Hanyang University
Spring & Fall
Supported various student initiatives at the College of Human Ecology throughout the year.
- 2019 Student Council (Department of Interior Architecture Design)** Hanyang University
Fall
Participated in the department's student council in order to directly support the activities of the Department of Interior Architecture and Design.